

Yunze Xiao

✉ lyxiao@cmu.edu | 🌐 algoxyolo | 🌐 yunzexiao

Education

Carnegie Mellon University

MS IN LANGUAGE TECHNOLOGY

Pittsburgh, PA

Aug. 2025 – Present

- Advisor: **Mona Diab** | GPA: 4.0/4.0

Carnegie Mellon University Qatar

BS IN COMPUTER SCIENCE

Doha, Qatar

Jul. 2022 – May 2025

- Advisor: **Houda Bouamor** | Minor in Computational Ethics
- Dean's List, 2025

Research Interests

I study how AI systems represent, interact with, and simulate people, from individual behavior to collective social dynamics. My work spans anthropomorphism and human-centered evaluation [P11], [P7]; persona and multi-agent social simulation [W4], [P2]; and the evidential foundations of AI welfare claims [P5]. Looking ahead, I am interested in scalable oversight and trustworthy multi-agent systems, with a focus on understanding and shaping their societal impact.

Research Experience

Carnegie Mellon University

ADVISOR: PROF. MONA DIAB

Pittsburgh, PA

Jan. 2025 – Present

- **Human-centered evaluation.** Co-authored frameworks for evaluating anthropomorphic design cues [P11] and cultural benchmarks [P7]; audited **20 cultural benchmarks** and identified six recurring methodological pitfalls.
- **Multi-agent social simulation.** Co-first author of *Sentipolis* [P2], which studies how persistent emotional states and memory shape agent interactions and emergent social dynamics.
- **Student simulation.** Co-authored a formalization of simulated learners as constrained generation under explicit knowledge-access, error, and learning-state specifications [W8].

Singapore University of Technology and Design

ADVISOR: PROF. ROY KA-WEI LEE

Singapore

May 2024 – Jul. 2024

- Developed **ToxiCloakCN**; implemented LLM and classifier baselines to measure macro-F1 degradation under homophone and emoji perturbations. EMNLP 2024, co-first author [P15].

Qatar Computing Research Institute

ADVISOR: DR. FIROJ ALAM

Doha, Qatar

May 2023 – Oct. 2023

- Fine-tuned Arabic BERT-based models, evaluated dropout settings, and benchmarked zero-shot and few-shot GPT-4 for propaganda and disinformation detection. ArAIEval 2023 [P18].

Carnegie Mellon University Qatar

ADVISOR: PROF. HOUDA BOUAMOR

Doha, Qatar

- **Multi-agent academic advising.** Co-authored *TartanMaroon* [P4], an advising system using iterative negotiation among specialized agents for constrained degree planning, evaluated on 220 advising queries. ACL 2026, System Demonstrations.
- **Chinese offensive language detection (May–Oct. 2023).** Led a survey of datasets, methods, and evaluation challenges [W11]; designed a labeling task and curated a **6k+ entry dataset** of subversive Chinese phrases.

Publications

* Equal contribution. Author order follows the original publications.

PUBLISHED

- [P1] Weihao Xuan, Qingcheng Zeng, Heli Qi, **Yunze Xiao**, Junjue Wang, and Naoto Yokoya. 2026. The Confidence Dichotomy: Analyzing and Mitigating Miscalibration in Tool-Use Agents. *ACL 2026*.
- [P2] Chiyuan Fu*, Lyuhao Chen*, **Yunze Xiao***, Weihao Xuan, Carlos Busso, and Mona Diab. 2026. Sentipolis: Emotion-Aware Agents for Social Simulations. *Findings of ACL 2026*.
- [P3] Ming Li*, Han Chen*, **Yunze Xiao**, Jian Chen, Hong Jiao, and Tianyi Zhou. 2026. Can LLMs Estimate Student Struggles? Human-AI Difficulty Alignment with Proficiency Simulation for Item Difficulty Prediction. *Findings of ACL 2026*.
- [P4] Peidi Dong, Houda Bouamor, **Yunze Xiao**, and Devi G Kurup. 2026. TartanMaroon: Multi-Agent Academic Advising with Iterative Negotiation and Transparent Collaboration. *ACL 2026, System Demonstrations*.
- [P5] **Yunze Xiao***, Gordon Dai*, Shahan Ali Memon*, Jen-tse Huang, Maarten Sap, and Mona T. Diab. 2026. Position: AI Welfare Is Bullshit. *ICML 2026, Position Paper Track*.
- [P6] **Yunze Xiao***, Tingyu He*, Lionel Z. Wang*, Yiming Ma, Xingyu Song, Xiaohang Xu, Mona Diab, Irene Li, and Ka Chung Ng. 2026. JiraiBench: A Cross-lingual Benchmark for Evaluating Large Language Models' Detection of Human Risky Health Behavior Content in Jirai Community. *EACL 2026. Oral*.
- [P7] Mai Alkhamissi*, **Yunze Xiao***, Badr Alkhamissi, and Mona Diab. 2026. Hire Your Anthropologist! Rethinking Culture Benchmarks Through an Anthropological Lens. *Findings of EACL 2026*.
- [P8] Lynnette Hui Xian Ng*, **Yunze Xiao***, Lionel Z. Wang, Weihao Xuan, and Mona Diab. 2026. ExpressionCueLens: a cross-cultural analysis of human-AI companion conversations on social media. *Journal of Ambient Intelligence and Humanized Computing*, 17, 1567–1578.
- [P9] Shu Yang, Junchao Wu, Xin Chen, **Yunze Xiao**, Xinyi Yang, Derek F. Wong, and Di Wang. 2026. Understanding Aha Moments: From External Observations to Internal Mechanisms. *TACL*.
- [P10] Center for AI Safety, Scale AI, and HLE Contributors Consortium (including **Yunze Xiao**). 2026. A benchmark of expert-level academic questions to assess AI capabilities. (Humanity's Last Exam). *Nature*, 649, 1139–1146.
- [P11] **Yunze Xiao***, Lynnette Hui Xian Ng*, Jiarui Liu, and Mona Diab. 2025. Humanizing Machines: Rethinking LLM Anthropomorphism Through a Multi-Level Framework of Design. *EMNLP 2025. Oral*.
- [P12] Weihao Xuan, Rui Yang, Heli Qi, Qingcheng Zeng, **Yunze Xiao**, Aosong Feng, Dairui Liu, Yun Xing, Junjue Wang, Fan Gao, Jinghui Lu, Yuang Jiang, Huitao Li, Xin Li, Kunyu Yu, Ruihai Dong, Shangding Gu, Yuekang Li, Xiaofei Xie, Felix Juefei-Xu, Foutse Khomh, Osamu Yoshie, Qingyu Chen, Douglas Teodoro, Nan Liu, Randy Goebel, Lei Ma, Edison Marrese-Taylor, Shijian Lu, Yusuke Iwasawa, Yutaka Matsuo, and Irene Li. 2025. MMLU-ProX: A Multilingual Benchmark for Advanced Large Language Model Evaluation. *EMNLP 2025*.
- [P13] Jiarui Liu, Yueqi Song, **Yunze Xiao**, Mingqian Zheng, Lindia Tjauatja, Jana Schaich Borg, Mona Diab, and Maarten Sap. 2025. Synthetic Socratic Debates: Examining Persona Effects on Moral Decision and Persuasion Dynamics. *EMNLP 2025*.
- [P14] Gordon Dai* and **Yunze Xiao***. 2025. Embracing Contradiction: Theoretical Inconsistency Will Not Impede the Road of Building Responsible AI Systems. *NeurIPS 2025, Position Paper Track*.
- [P15] **Yunze Xiao***, Yujia Hu*, Kenny Tsu Wei Choo, and Roy Ka-Wei Lee. 2024. ToxiCloakCN: Evaluating Robustness of Offensive Language Detection in Chinese with Cloaking Perturbations. *EMNLP 2024*.
- [P16] Xintao Wang, **Yunze Xiao**, Jen-tse Huang, Siyu Yuan, Rui Xu, Haoran Guo, Quan Tu, Yaying Fei, Ziang Leng, Wei Wang, Jiangjie Chen, Cheng Li, and Yanghua Xiao. 2024. InCharacter: Evaluating Personality Fidelity in Role-Playing Agents through Psychological Interviews. *ACL 2024*.
- [P17] David R. Mortensen*, Valentina Izrailevitch*, **Yunze Xiao**, Hinrich Schütze, and Leonie Weissweiler. 2024. Verbing Weirds Language (Models): Evaluation of English Zero-Derivation in Five LLMs. *LREC-COLING 2024*.
- [P18] **Yunze Xiao** and Firoj Alam. 2023. Nexus at ArAIEval Shared Task: Fine-Tuning Arabic Language Models for Propaganda and Disinformation Detection. *ArabicNLP 2023*.
- [P19] **Yunze Xiao**. 2022. A Transformer-based Attention Flow Model for Intelligent Question and Answering Chatbot. *ICCRD 2022*, 167–170.

PREPRINTS

- [W1] Haokai Zhao, **Yunze Xiao**, Weihao Xuan, Flora Salim, Benjamin Tag, and Aditya Joshi. 2026. Group Alignment-Induced Sycophancy: A Two-Sided Evaluation of Steerable Pluralistic Alignment. *arXiv:2608.11528*.
- [W2] Yueqi Song, Lintang Sutawika, Jiarui Liu, Lindia Tjautja, Jiayi Geng, **Yunze Xiao**, Daniel Lee, Aditya Bharat Soni, Vincent Lo, Xiang Yue, and Graham Neubig. 2026. PACE: A Proxy for Agentic Capability Evaluation. *arXiv:2607.02032*.
- [W3] Weijia Zhang, Ruiqi Chen, **Yunze Xiao**, and Weihao Xuan. 2026. Every Act Has Its Price: Compressed Moral Composition in Frontier LLMs. *arXiv:2606.11232*.
- [W4] **Yunze Xiao**^{*}, Vivienne J. Zhang^{*}, Chenghao Yang, Ningshan Ma, Weihao Xuan, and Jen-tse Huang. 2026. The Chameleon’s Limit: Investigating Persona Collapse and Homogenization in Large Language Models. *arXiv:2604.24698*.
- [W5] Xirui Li, Ming Li, **Yunze Xiao**, Ryan Wong, Dianqi Li, Timothy Baldwin, and Tianyi Zhou. 2026. Superminds Test: Actively Evaluating Collective Intelligence of Agent Society via Probing Agents. *arXiv:2604.22452*.
- [W6] **Yunze Xiao**, Wenkai Li, Xiaoyuan Wu, Ningshan Ma, Yueqi Song, and Weihao Xuan. 2026. Say Something Else: Rethinking Contextual Privacy as Information Sufficiency. *arXiv:2604.06409*.
- [W7] Xuan Liu, Haoyang Shang, Zizhang Liu, Xinyan Liu, **Yunze Xiao**, Yiwen Tu, and Haojian Jin. 2026. HumanStudy-Bench: Towards AI Agent Design for Participant Simulation. *arXiv:2602.00685*.
- [W8] Zhihao Yuan^{*}, **Yunze Xiao**^{*}, Ming Li^{*}, Weihao Xuan, Richard Tong, Mona Diab, and Tom Mitchell. 2026. Towards Valid Student Simulation with Large Language Models. *arXiv:2601.05473*.
- [W9] Rui Yang, Huitao Li, Weihao Xuan, Heli Qi, Xin Li, Kunyu Yu, Yingjian Chen, Rongrong Wang, Jacques Behmoaras, Tianxi Cai, Bibhas Chakraborty, Qingyu Chen, Lionel Tim-Ee Cheng, Marie-Louise Damwanza, Chido Dzinotyiwei, Aosong Feng, Chuan Hong, Yusuke Iwasawa, Yuhe Ke, Linah Kitale, Taehoon Ko, Jisan Lee, Irene Li, Jonathan Chong Kai Liew, Hongfang Liu, Lian Leng Low, Edison Marrese-Taylor, Yutaka Matsuo, Isheanesu Misi, Yilin Ning, Jasmine Chiat Ling Ong, Marcus Eng Hock Ong, Enrico Petretto, Hossein Rouhizadeh, Abiram Sandrelegar, Oren Schreier, Iain Bee Huat Tan, Patrick Tan, Daniel Shu Wei Ting, Junjue Wang, Chunhua Weng, Matthew Yu Heng Wong, Fang Wu, **Yunze Xiao**, Xuhai Xu, Qingcheng Zeng, Zhuo Zheng, Yifan Peng, Douglas Teodoro, and Nan Liu. 2026. Toward Global Large Language Models in Medicine. *arXiv:2601.02186*.
- [W10] Qingyang Wu, Ying Xu, Tingsong Xiao, **Yunze Xiao**, Yitong Li, Tianyang Wang, Yichi Zhang, Shanghai Zhong, Yuwei Zhang, Wei Lu, and Yifan Yang. 2024. Surveying Attitudinal Alignment Between Large Language Models Vs. Humans Towards 17 Sustainable Development Goals. *arXiv:2404.13885*.
- [W11] **Yunze Xiao**, Houda Bouamor, and Wajdi Zaghouani. 2024. Chinese Offensive Language Detection: Current Status and Future Directions. *arXiv:2403.18314*.

Academic Service

ACADEMIC LEADERSHIP

- 2026 ACL Rolling Review, Area Chair
- 2026 SocialSim @ COLM, Organizer
- 2025 PersonaNLP @ NeurIPS, Lead Organizer

PROGRAM COMMITTEES & REVIEWING

- 2025 AAAI Student Abstract and Poster Program, Program Committee Member

Conferences: ACL (2025, 2026); CAIS (2026); CSCW (2025); EMNLP (2025); ICASSP (2026); ICLR (2025, 2026); ICWSM (2025); WWW (2026).

Journals: IJHCI; TAFFC; TMLR; TNNLS.

UNIVERSITY & COMMUNITY SERVICE

- 2026 LTI Teaching Track Search Committee, Student Member
- 2024 The Web Conference, Volunteer
- 2023 Student Majilis, Head of Academics

Talks & Presentations

Persona Collapse: Measuring Diversity Failures in LLM-Simulated Populations. STAMINA-WG Research Talk Series on Agents, Sims, Social Tech, and Safety. *May 2026.*

Embracing Contradiction: Theoretical Inconsistency Will Not Impede the Road of Building Responsible AI Systems. CMU FEAT Reading Group. *Feb. 2026.*

Humanizing Machines: Rethinking LLM Anthropomorphism Through a Multi-Level Framework of Design. MLNLP Seminar. *Oct. 2025.*

Classical and Modern Language Modeling Techniques for Transmembrane Helix Prediction. ISMB, extended abstract poster. With Malak Ibrahim and Madhavi Ganapathiraju. *2024.*

Research Mentoring

Peidi Dong (CMU Qatar, undergraduate): AI safety, HAI, education [P4].

Sep. 2025 – Present

Ethan Pan (UW-Madison, undergraduate): Social simulation, HCI.

Sep. 2024 – Present

Teaching

- Spr. 2024 98-031: NLP Ethics in a Nutshell, Student Instructor
- Spr. 2024 15-112: Fundamentals of Programming and Computer Science, Course Assistant
- Fall 2023 15-110: Principles of Computing, Course Assistant